

ADOLESCENT SUICIDAL IDEATION PREDICTION: INTEGRATING STATISTICAL AND MACHINE LEARNING APPROACHES

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Introduction

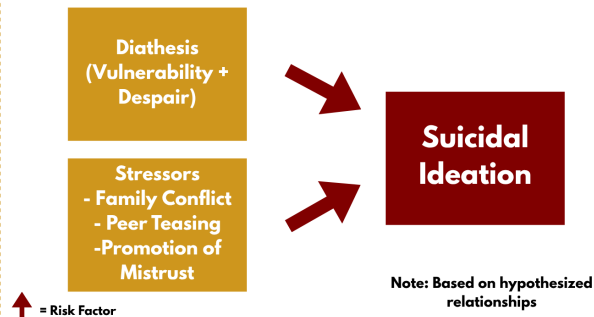
- **Suicide** remains a **leading cause of death** for **adolescents** globally (United Nations Children's Fund, 2021). Prior research has identified several risk factors for adolescent suicidal ideation, including psychiatric problems, family conflict, parental rejection, and peer victimization and bullying (Steinberg, 2020).

Diathesis-Stress Model: Suicidal ideation emerges when pre-existing vulnerability (diathesis) combines with environmental stressors.

(Broerman, 2020; Monroe & Simons, 1991)

- **Research Gap:** empirical evidence on mental health risk factors among adolescents, particularly for ethnic-racial minority youth in the Majority World, remains limited (United Nations Children's Fund, 2021).
- **Purpose & Hypothesis:** This study investigated the impact of despair, parent-child conflict, gender-based peer teasing, and parental promotion of mistrust on suicidal ideation among Korean-Chinese adolescents in Northeast China. We hypothesized that these factors would positively predict suicidal ideation.

Diathesis-Stress Model of Suicidal Ideation



References

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Methods

Participants: N = 267 Korean-Chinese adolescents from bilingual secondary schools in Northeast China.

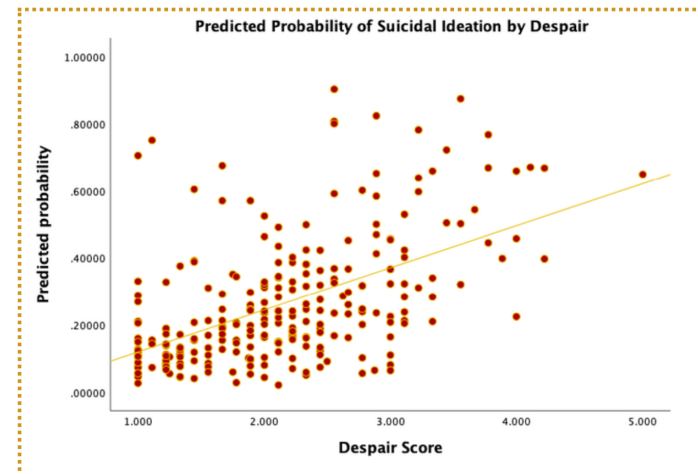
Measures

- Suicidal Ideation: single yes/no item.
 - ex: During the past 12 months, did you ever seriously think about committing suicide?
- Despair ($\alpha = 0.905$): despair-only subscale of the Beck Hopelessness Scale (Beck, 1974). 11 items rated on a likert scale.
 - ex: Because I cannot make my life better, it may be better to give up.
- Parent-Child Conflict ($\alpha = 0.727$): 4 items adapted from the Conflict Behavior Questionnaire
 - ex: My mom and I get angry at each other.
- Gender-Based Teasing: single item.
 - ex: Kids at school tease me for behaving like a girl when I'm a boy and vice versa.
- Gender Role ($\alpha = 0.740$): 7 items assessing parental traditional gender beliefs.
 - ex: My parents think that girls should not date while in high school.
- Promotion of Mistrust ($\alpha = 0.772$): 3 items from Tran & Lee (2010).
 - ex: Done or said things to encourage you to keep distance from people of other ethnicities?
- Socioeconomic Status: single self-report item.
 - ex: If you are asked about your family's socio-economic status, which of the following classes would you say your family belongs to?

Dual-Method Approach

- 1 **Inferential Statistics:** Binary logistic regression conducted in IBM SPSS. Outcome variable coded as binary (0 = no suicidal ideation, 1 = suicidal ideation).
- 2 **Machine Learning:** Logistic regression model[‡] implemented in Python/Pandas with a 70/30 train-test split.

[‡]Model development used Gemini-assisted coding and DataCamp modeling frameworks.



$$P(\text{SuicidalIdeation}) = \frac{e^{-6.141+0.771(\text{Despair})+0.543(\text{Conflict})+0.514(\text{Teasing})-0.913(\text{Mistrust})}}{1 + e^{-6.141+0.771(\text{Despair})+0.543(\text{Conflict})+0.514(\text{Teasing})-0.913(\text{Mistrust})}}$$

Statistical Results

The model identified several **significant predictors** of suicidal ideation:

- **Despair:** OR = 2.16, $p < .001$.
- **Parent-Child Conflict:** OR = 1.72, $p = .009$.
- **Gender-Based Teasing:** OR = 1.67, $p = .001$.
- **Promotion of Mistrust:** OR = 0.40, $p = .008$ (Significant negative predictor).
- Non-Significant Predictors: Gender, socioeconomic status, bicultural identity, school social support

Machine Learning Results

- The model was evaluated on its ability to classify **at-risk** adolescents:
 - Accuracy: **55%** (Overall correct predictions).
 - Recall: **64%** (Ability to correctly identify those actually experiencing suicidal ideation).

Discussion

- Consistent with the **diathesis-stress model**, despair emerged as the strongest predictor of suicidal ideation (OR = 2.16). Environmental stressors such as parent-child conflict (OR = 1.72) and gender-based peer teasing (OR = 1.67) significantly elevated risk. Surprisingly, promotion of mistrust negatively predicted suicidal ideation (OR = 0.40); among Asian youth, higher promotion of mistrust has been linked to lower social competence, suggesting its effects on wellbeing may be population-specific (Tran & Lee, 2010).
- **Impact & Interventions:** Suicide ranks among the top causes of death for adolescents aged 15-19 globally (UNICEF, 2021). Interventions targeting despair may benefit from approaches addressing hopelessness and self-blame (Steinberg, 2020). Addressing family conflict and gender-based peer teasing through school and family support programs aligns with Bronfenbrenner's emphasis on strengthening protective microsystems (Bronfenbrenner & Morris, 2006).
- **Limitations & Future Research:** This study's sample was drawn from two schools in Northeast China; future research should use larger datasets to validate findings, improve machine learning model accuracy, and further examine promotion of mistrust.